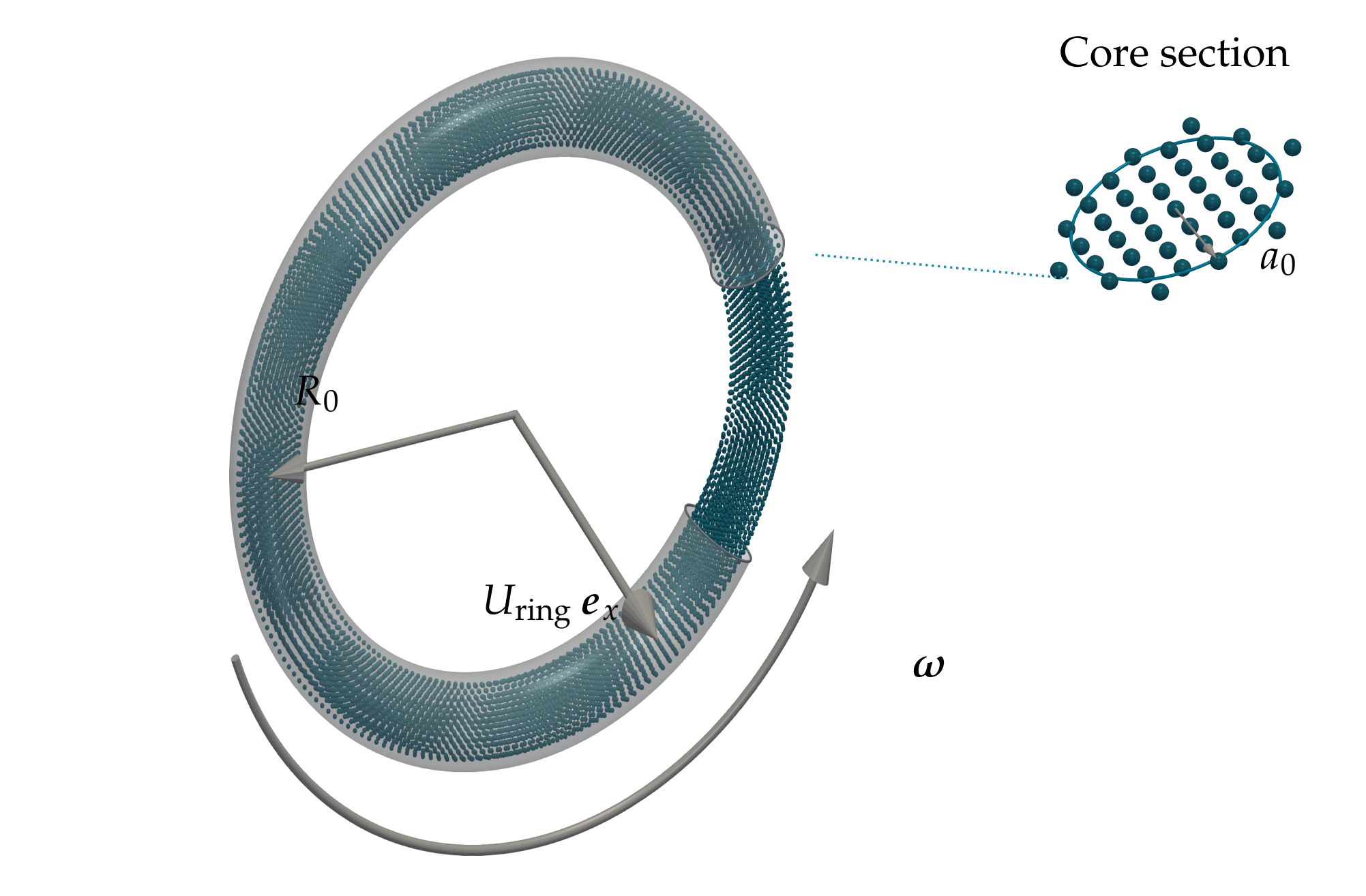

Core section
a_0
R_0
$U_{\text{ring}}\, \boldsymbol{e}_x$
$\boldsymbol{\omega}$